

ENTRY 039 790 Circuits: the Estimates Are Converging — August 16, 2026 — Status: COMPLETE

Entry 039 — A Fourth Independent Batch, and the Numbers Stop Moving

Balanced Across All Six Hop-Distance Bins Again, Restoring the Long-Route Coverage Entry 038 Traded Away

1. Why Balance This Time

Entry 038 deliberately skewed toward short/mid routes to grow the floor-collapse sample cheaply, which worked, but left the dataset thin on long routes again. entry039_grow_balanced.py returns to all six hop-distance bins from Entries 034/035 (with a fourth independent seed) plus a GHZ batch, adding 232 more circuits: 232 new (117 Kyiv, 115 Sherbrooke), on top of the 558 from Entries 034/035/038.

2. The Floor-Collapse Ratio Holds Steady

	after Entry 038 (n=396 short-route)	after Entry 039 (n=535 short-route)
total circuits	558	790
floor-collapse cases	37	47
Kyiv short-route rate	6.1%	6.0%
Sherbrooke short-route rate	12.5%	11.6%
Sherbrooke / Kyiv ratio	2.05x	1.93x

Two independent growth rounds in a row now land within a few tenths of a point of each other -- 2.05x, then 1.93x. That's the signature of an estimate that has actually converged rather than one still drifting with more data, in contrast to the 4.7x-to-2.05x jump between the original small sample and the first growth round. Sherbrooke floor-collapsing on short routes roughly twice as often as Kyiv looks like a stable, real property of the two chips.

3. The ML Baseline's Advantage Also Holds

	after Entry 038 (37 fc cases)	after Entry 039 (47 fc cases)
v4.1 floor-collapse MAE	18.20 pts	17.59 pts
Gradient boosting floor-collapse MAE	14.79 pts (-19%)	14.07 pts (-20%)
Random forest floor-collapse MAE	15.34 pts (-16%)	15.70 pts (-11%)

Gradient boosting's relative advantage over v4.1 on floor-collapse cases holds essentially flat (-19% to -20%) across two growth rounds. Random forest's edge narrowed slightly. Neither model is closing the gap further as data grows within this range -- consistent with Entry 036's read that the missing ingredient is a correlated-noise feature these models don't have access to, not simply more of the same kind of data.

Entry 039 — Conclusion

Two consecutive independent growth rounds (558 to 790 circuits) now agree closely on both headline numbers: Sherbrooke's short-route floor-collapse rate is roughly 2x Kyiv's, and a tabular ML model's edge over v4.1 on those cases plateaus around 15-20% relative MAE reduction. That plateau is itself informative -- it suggests the current feature set has been mostly saturated, and further gains will need either a genuinely new feature (something that captures cross-channel correlation directly) or the GNN architecture originally planned, once there is enough data to support one.

4. Where This Leaves the Project

790 circuits is a real, useful dataset -- roughly 79x the original 5-circuit reference sets this project started validating against in Entry 022 -- but still short of what a GNN needs, and the plateau in section 3 suggests raw volume alone won't move the floor-collapse number much further without a new feature or architecture. This is a reasonable point to pause pure data growth and let the next entry decide between two paths: keep scaling the existing feature set toward

a GNN, or first engineer a feature that captures the specific correlated-noise signature Entry 036 found.

Research Log — Index Addendum (Entry 039)

Entry	Title	Date	Status
Entry 039	Fourth Growth Batch (790 circuits total); Estimates Converge	Aug 16, 2026	Complete

Script: entry039_grow_balanced.py. Data: quantumbridge_data/entry039_balanced_dataset.json (232 new), quantumbridge_data/entry039_combined_dataset.json (790 total, merged), quantumbridge_data/entry039_growth_summary.json. Floor-collapse ratio and ML advantage both stabilized across two growth rounds -- a plateau, not further movement.